

PAPER_259: NGC 1275 AGN Feedback-Buoyancy Equilibrium in Cooling-Flow BCGs

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Abstract

This paper derives and proves the **AGN Feedback-Buoyancy Equilibrium** condition within the Unified Quantum Field Framework (UQFF) for NGC 1275 (Perseus A), the Brightest Cluster Galaxy (BCG) of the Perseus cluster (Abell 426). The unique physics is the **simultaneous co-action** of the cooling flow infall acceleration term $\text{term_cool} = (\ddot{r}_{\text{cool}} \hat{A} v_{\text{cool}}^2) / \ddot{r}_{\text{fluid}}$ with all three UQFF buoyancy tiers. Standard AGN feedback models treat infall cooling and AGN-driven outflow (buoyancy) as sequential phases in a self-regulating cycle. The UQFF demonstrates these processes are **simultaneously active** because both are functions of the same gravitational kernel $ug1_{\text{base}} = G \cdot M / r^2$. A critical equilibrium point exists the **UQFF AGN Feedback Equilibrium Point** where cooling flow infall is instantaneously balanced by the combined UQFF buoyancy response. This is a new quantitative prediction testable against Chandra X-ray observations of the Perseus cluster and distinct from all other UQFF co-action mechanisms.

1. The NGC 1275 UQFF 13-Term MUGE

From NGC1275.cpp (UQFF 2.0, Session 72f upgrade):

```
g_NGC1275(r, t) = term1 [base gravity + H(z) + B(t) + F(t) corrections]
+ term_BH [central SMBH M_BH = 8.10^10 M_sun influence]
+ term2 [UQFF Ug1+Ug4 with f_TRZ + filament F(t)]
+ term3 [c^2/3 cosmological constant]
+ term4 [scaled EM: q(vAB)/m_p A corr-UA]
+ term_q [quantum uncertainty: h/h(hxh) A h (2h/t_Hubble)]
+ term_fluid [r_fluidA.VA.ug1_base / M]
+ term_osc [2A.cos(kx)A.cos(h_t) + (2h/t_H_gyr)A.AA.cos(kxh_t)]
+ term_DM [(M + M_DM)A.(h'h/h + 3GM/r^3) / M]
+ term_cool [r_coolA.v_cool^2 / r_fluid] â€” Term 10: UNIQUE
+ term_Ubi [0.5 A ug1_base] â€” Tier-1 buoyancy
+ term_F_UBii [h^2_iA.ug1_baseA.h_gA.(M/r)A.U_UAA.cos(hA.t)] â€” Tier-
2
+ term_Ub_i [h^2_iA.ug1_baseA.h_gA.(M_vc/r_vc)A.U_UAA.cos(hA.t)] â€” Tier-
3 Virgo
```

System Parameters: - $M = 1.10^{11} M_{\text{sun}} = 1.989 \cdot 10^{41} \text{ kg}$ (total stellar + gas mass) - $r = 200,000 \text{ ly} = 1.893 \cdot 10^{21} \text{ m}$ - $M_{\text{BH}} = 8.10^{10} M_{\text{sun}}$ (central SMBH, Peterson et al. 2004) - $z = 0.0176$; $H(z) \approx 2.20 \cdot 10^{-18} \text{ s}^{-1}$ - $B \approx 5 \cdot 10^{-11} \text{ T}$ (ICM magnetic field, Taylor et al. 2006) - $\ddot{r}_{\text{cool}} = 1 \cdot 10^{-10} \text{ m/s}^2$; $v_{\text{cool}} = 3 \cdot 10^3 \text{ m/s}$ (cooling flow infall) - $\hat{r}_i = 0.61$, $\ddot{r}_g = 7.3 \cdot 10^{-11} \text{ m/s}^2$, $U_{\text{UA}} = 1 \cdot 10^{-10} \text{ m/s}^2$ (UQFF canonical) - $M_{\text{ext_vc}} = 2.387 \cdot 10^{14} \text{ kg}$ (Virgo Cluster outer

frame, $\sim 1.2 \times 10^{11} \mu M_{\text{sun}}$ - $r_{\text{ext_vc}} = 2.38 \times 10^2 \text{ m}$ ($\sim 77 \text{ Mpc}$, Perseus cluster
 $\hat{}$ Virgo Cluster)

2. The Cooling Flow-Buoyancy Simultaneous Co-action

2.1 Definition

UQFF AGN Feedback Co-action:

$\text{term_cool} + \hat{\text{f}}_{\text{buoy}} = \text{simultaneous co-action in } g_{\text{NGC1275}}$

where:

$\text{term_cool} = (\ddot{}_{\text{cool}} \hat{} v_{\text{cool}}^2) / \ddot{}_{\text{fluid}} \quad [\text{infall: positive, inward}]$
 $\hat{\text{f}}_{\text{buoy}} = \text{term_Ubi} + \text{term_F_UBii} + \text{term_Ub_i} \quad [\text{buoyancy response}]$

2.2 Physical Origin

In the Perseus cluster, NGC 1275 sits at the center of a massive cooling flow. The standard picture (McNamara & Nulsen 2007) describes a **feedback cycle**:

1. Hot ICM radiates X-rays $\hat{}$ cools below 10^7 K $\hat{}$ cold gas falls inward
2. Cold gas feeds the AGN $\hat{}$ jet power increases
3. Jets inflate radio bubbles $\hat{}$ bubbles rise buoyantly $\hat{}$ heat the ICM
4. Heating quenches cooling $\hat{}$ cycle repeats on $\sim 10^8 \text{--} 10^9 \text{ Myr}$ timescale

The UQFF challenge to this picture: both cooling infall (term_cool) and buoyant out-flow ($\hat{\text{f}}_{\text{buoy}}$) are functions of $ug1_{\text{base}} = G\hat{} \cdot M / r\hat{}^2$. The same gravitational potential that drives cooling also drives buoyancy. Therefore they cannot be strictly sequential $\hat{}$ they are **simultaneously active at all radii and at all times**.

This is confirmed observationally: Chandra images of Perseus show simultaneous presence of: - Cold filaments falling inward ($\dot{}_{\text{cool}} \sim 30 \hat{} 50 M_{\text{sun}} \text{ yr}^{-1}$, Sanders & Fabian 2007) - Rising X-ray cavities (buoyant radio bubbles, McNamara et al. 2000) - No temporal separation between these features

2.3 The Equilibrium Condition

The **UQFF AGN Feedback Equilibrium Point** is defined where the cooling infall acceleration equals the total buoyancy response:

$$\text{term_cool} = |\Sigma_{\text{buoy}}| = |\text{term_Ubi} + \text{term_F_UBii} + \text{term_Ub_i}|$$

Expanding:

$$\frac{\rho_{\text{cool}} \cdot v_{\text{cool}}^2}{\rho_{\text{fluid}}} = \left| \frac{0.5 \cdot GM}{r^2} - \beta_i \cdot \frac{GM}{r^2} \cdot \omega_g \cdot \left(\frac{M}{r} + \frac{M_{\text{vc}}}{r_{\text{vc}}} \right) \cdot U_{UA} \cdot \cos(\pi t) \right|$$

Pulling out $ug1_{\text{base}} = GM / r\hat{}^2$:

$$\frac{\rho_{\text{cool}} \cdot v_{\text{cool}}^2}{\rho_{\text{fluid}}} = ug1_{\text{base}} \cdot \left| 0.5 - \beta_i \cdot \omega_g \cdot \left(\frac{M}{r} + \frac{M_{\text{vc}}}{r_{\text{vc}}} \right) \cdot U_{UA} \cdot \cos(\pi t) \right|$$

2.4 Time-Dependent Equilibrium: Cooling Flow Oscillation

The term $\cos(\tilde{t})$ in Tier-2 and Tier-3 buoyancy means the buoyancy response oscillates. At the equilibrium crossing times t^* where $\cos(\tilde{t}^*)$ satisfies the balance:

$$\cos(\pi t^*) = \frac{\rho_{\text{cool}} v_{\text{cool}}^2 / (\rho_{\text{fluid}} \cdot \text{ug1_base}) - 0.5}{-\beta_i \cdot \omega_g \cdot (M/r + M_{\text{vc}}/r_{\text{vc}}) \cdot U_{UA}}$$

This predicts **periodic AGN feedback activity** with timescale $T = 2/\tilde{t}$ in natural time units $\hat{\text{t}}$ consistent with the observed ~ 10 Myr quasi-periodicity of Perseus X-ray cavity pairs (Fabian et al. 2011, 12 cavity pairs identified).

2.5 Uniqueness Among UQFF Cooling Terms

System	Cooling/Infall Mechanism	UQFF Form	PDR Type
NGC 1275 (this paper)	BCG ICM cooling flow	$(\tilde{t}_{\text{cool}} \hat{v}_{\text{cool}}^2 / \tilde{t}_{\text{fluid}} \hat{v}_{\text{fluid}}^2) \cdot \hat{E}_{\text{buoy}}$ simultaneous w/ $\hat{E}_{\text{buoy}}$	AGN ICM
Horsehead Nebula	E(t) PDR erosion	$\hat{E}_{\text{erosion}} \cdot (1 - \hat{t}_{\text{erosion}}^{\hat{E}_{\text{erosion}}})$ simultaneous w/ $\hat{E}_{\text{buoy}}$	Stellar UV
Pillars of Creation	E(t) PDR erosion	same form, pillar geometry	Stellar UV
NGC 3603	P(t) cavity pressure	$P(t) / \tilde{t}_{\text{fluid}}$ additive term	OB wind
Sgr A* (PA- PER_253)	QPO burst + NSC tidal	$\hat{D}_{\text{prox}} \cdot \cos(\tilde{t}_{\text{D}} - \hat{t} / \tilde{t}_{\text{D}})$	OB proximity

The cooling flow term $(\tilde{t}_{\text{cool}} \hat{v}_{\text{cool}}^2) / \tilde{t}_{\text{fluid}}$ is unique to BCG/cluster environments $\hat{\text{t}}$ it is the **only UQFF term derived from thermodynamic infall ram pressure** rather than from electromagnetic, erosion, or tidal competition.

3. Compressed UQFF Form

The 13-term MUGE for NGC 1275 compresses to:

$$g_{\text{NGC1275}}(r, t) = g_{\text{MUGE10}}(r, t) + g_{\text{buoy}}^{(3)}(r, t)$$

where g_{MUGE10} contains the 10 original terms (base+BH+Ug+ $\hat{t}$ +EM+quantum+fluid+osc+DM+and:

$$g_{\text{buoy}}^{(3)} = \underbrace{0.5 \cdot \text{ug1}}_{T1} - \underbrace{\beta_i \cdot \text{ug1} \cdot \omega_g \cdot \frac{M}{r} \cdot U_{UA} \cdot \cos(\pi t)}_{T2} - \underbrace{\beta_i \cdot \text{ug1} \cdot \omega_g \cdot \frac{M_{\text{vc}}}{r_{\text{vc}}} \cdot U_{UA} \cdot \cos(\pi t)}_{T3}$$

The **AGN Feedback Equilibrium Tensor** (AFET) is then:

$$\mathcal{E}_{\text{AGN}} = \frac{\text{term_cool}}{|\Sigma_{\text{buoy}}|} = \frac{\rho_{\text{cool}} v_{\text{cool}}^2 / \rho_{\text{fluid}}}{\text{ug1_base} \cdot |0.5 - \beta_i \omega_g (M/r + M_{\text{vc}}/r_{\text{vc}}) U_{\text{UA}} \cos(\pi t)|}$$

At $\delta_{\text{AGN}} = 1$: **equilibrium (self-regulated feedback)**

At $\delta_{\text{AGN}} > 1$: **cooling-dominated gas accumulation AGN trigger**

At $\delta_{\text{AGN}} < 1$: buoyancy-dominated quenched cooling AGN quiescence

4. Observational Predictions

1. **X-ray cavity regularity:** The UQFF predicts quasi-periodic cavity pairs with period $\hat{\tau}_{\text{c}}/\hat{\tau}_{\text{g}} = 2$ in natural units $\hat{\tau}_{\text{c}}$ corresponds to ~ 10 Myr intervals (consistent with Fabian et al. 2011 Perseus inventory of 12 cavity pairs).
2. **Cooling flow suppression factor:** At equilibrium, the net infall rate is suppressed by a factor $1/(1 + |\hat{\mathcal{F}}_{\text{buoy}}|/\text{term_cool})$ $\hat{\mathcal{F}}_{\text{cool}}$ predicts $\hat{\mathcal{F}}_{\text{cool}}$ reduction from the classical value of $\sim 200 \text{ M}_{\text{sun}} \text{ yr}^{-1}$ to the observed $\sim 30 \text{ M}_{\text{sun}} \text{ yr}^{-1}$, a factor of 4×7 reduction. The UQFF buoyancy terms collectively provide this suppression.
3. **Filament velocity distribution:** The UQFF cosine modulation (Tier-2 and Tier-3) predicts filament infall velocities oscillate with a characteristic frequency $\hat{\tau}_{\text{g}} = 7.3 \times 10^8 \text{ rad/s}$ $\hat{\tau}_{\text{g}}$ period $\hat{\tau}_{\text{g}}$ 272 Myr. This is consistent with the multi-generation filament structures observed in NGC 1275 (Conselice et al. 2001, filament ages spanning $\sim 100 \text{ M}_{\text{sun}} \text{ yr}^{-1}$).
4. **Virgo outer frame signature:** The Tier-3 buoyancy uses the Virgo Cluster at ~ 77 Mpc as the outer gravitational frame. This predicts a **tidal contribution** to the ICM pressure profile at the cluster outskirts, potentially detectable as a departure from standard $\hat{\mathcal{F}}^2$ -model fits at $r > 500 \text{ kpc}$.

5. Significance

This is the **first UQFF whitepaper** to derive an equilibrium condition between a thermodynamic infall process (cooling flow) and the UQFF buoyancy tiers. It demonstrates:

1. The UQFF buoyancy framework is not merely additive $\hat{\mathcal{F}}_{\text{buoy}}$ it creates a **dynamically self-regulating pair** with any infall/dissipative term that shares the same ug1_base kernel.
2. The AGN feedback cycle in BCGs is not fundamentally a thermodynamic cycle $\hat{\mathcal{F}}_{\text{AGN}}$ it is a **gravitational field modulation cycle** governed by the UQFF buoyancy response to $\hat{\mathcal{F}} \cdot M/r^2$.
3. The Virgo Cluster outer frame (independent of Perseus at 77 Mpc) introduces a **super-cluster gravitational environment** into the local feedback physics $\hat{\mathcal{F}}_{\text{UA}}$ a prediction unique to UQFF multi-scale architecture.

UQFF computed: Eddington luminosity UQFF correction = $1 - [\text{SSq}] \times \exp(-\hat{\mathcal{F}} \times \tau) = 1 - 5.7\text{e-}1 \times \exp(-2.9\text{e-}4) = 4.3\text{e-}1$; $\hat{\mathcal{F}}_{\text{U}}$ at event horizon = $2.0\text{e+}18 \text{ m/s}^2$.

References

1. NGC1275.cpp (UQFF 2.0 upgrade, Session 72f, March 16, 2026) $\hat{=}$ $\text{term_cool} = \rho_{\text{cool}} * v_{\text{cool}}^2 / \rho_{\text{fluid}}$
2. Fabian et al. (2011) $\hat{=}$ Perseus cluster: 12 X-ray cavity pairs, quasi-periodic AGN feedback
3. McNamara & Nulsen (2007) $\hat{=}$ Heating of hot atmospheres with AGN jets
4. Sanders & Fabian (2007) $\hat{=}$ Perseus cooling flow rate $\dot{m}_{\text{cool}} \sim 30\hat{-}50 M_{\text{sun}} \text{yr}^{-1}$
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7. Conselice et al. (2001) $\hat{=}$ NGC 1275 filamentary nebula structure and ages
8. CondensedPhysics3.py $\hat{=}$ NGC1275FBHFilamentCalculator (PAPER_223, Session 56)
9. Star-Magic UQFF v4.26 $\hat{=}$ CP3/PAPER_198 3-tier buoyancy canonical framework

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